



SCEGGS Darlinghurst

2005
Higher School Certificate
Assessment Task 2
Friday 10th June

Mathematics Extension 1

Task Weighting: 35%

General Instructions

- Time allowed – 75 minutes
- This paper has **four** questions
- Attempt **all** questions
- Write using blue or black pen
- Answer all questions on the pad paper provided
- Begin each question **on a new page**
- Write your name at the top of each page
- Mathematical templates, geometrical equipment and approved scientific calculators may be used
- A table of standard integrals is provided

Name:

Teacher:

Question	Comm	Reason	Calculus	Marks
1	/4		/3	/15
2		/7	/7	/15
3		/3	/3	/16
4	/3	/5	/4	/14
TOTAL	/7	/15	/17	/60

Question 1 (15 marks)

Marks

- (a) The interval AB lies between $A(-1, 4)$ and $B(5, -3)$. Find the co-ordinates of the point P which divides the interval AB externally in the ratio 1:3. **2**
- (b) Find $\int \frac{x}{(x^2 + 2)^2} dx$ using the substitution $u = x^2 + 2$. **3**
- (c) (i) State the domain and range of $y = 3\sin^{-1} 2x$. **2**
- (ii) Sketch $y = 3\sin^{-1} 2x$, showing all important features. **1**
- (d) Solve the inequality $\frac{2x}{x-2} \leq 3$. **3**
- (e) (i) Sketch $f(x) = \sqrt{x-2}$. **1**
- (ii) Explain why the function $f(x) = \sqrt{x-2}$ has an inverse function $f^{-1}(x)$. **1**
- (iii) Write down the equation of the inverse function $f^{-1}(x)$. **1**
- (iv) On the same set of axes as your graph for (i), sketch $y = f^{-1}(x)$. **1**

Question 2 (15 marks)

Marks

- (a) Find the exact value of $\tan\left(\sin^{-1}\left(-\frac{1}{3}\right)\right)$. 2
- (b) (i) Sketch the curve $y = 1 + \cos x$ for $0 \leq x \leq 2\pi$. 1
- (ii) Calculate the exact volume generated when the arc of $y = 1 + \cos x$ between $x = 0$ and $x = \frac{3\pi}{2}$ is rotated about the x -axis. 3
- (c) Use the substitution $u = 1 - x$ to find $\int_0^1 \frac{x}{\sqrt{1-x}} dx$. 4
- (d) (i) Find $\frac{d}{dx} \left[\sqrt{1-x^2} + x \sin^{-1} x \right]$. 3
- (ii) Hence, evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x dx$. 2
Leave your answer in exact form.

Question 3 (16 marks)

Marks

- (a) (i) Find the exact gradients of the tangents to the curve $y = \sin^{-1} \frac{x}{2}$ at $x = 0$ and $x = 1$. 3
- (ii) Find the acute angle between these two tangents correct to the nearest minute. 2
- (b) (i) Show that $3 \sin \theta + \cos \theta = 2$ can be written as $3t^2 - 6t + 1 = 0$ if $t = \tan \frac{\theta}{2}$. 2
- (ii) Hence, or otherwise, solve $3 \sin \theta + \cos \theta = 2$ for $0^\circ \leq \theta \leq 360^\circ$. 3
Leave your answers correct to the nearest minute.
- (c) Show that $\cos^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{-3}{4}\right) = \frac{\pi}{2}$. 3
- (d) Evaluate $\int_0^{\frac{5}{2}} \frac{dx}{\sqrt{25-2x^2}}$. 3

Question 4 (14 marks)

Marks

- (a) Find the general solution/s in radians for which $3\cos 2x = 2 + \sin x$.

3

- (b) Use the substitution $u = \sin x$ to show $\int_0^{\pi/6} \frac{\cos x}{4\sin^2 x + 1} dx = \frac{\pi}{8}$.

4

- (c) (i) State the domain and range of $y = 2\cos^{-1}(1-x)$.

2

- (ii) Sketch the curve $y = 2\cos^{-1}(1-x)$.

1

- (iii) On the same set of axes, sketch $y = -\pi x + 2\pi$.

1

- (iv) Explain why:

1

$$\int_0^2 2\cos^{-1}(1-x) dx = \int_0^2 (-\pi x + 2\pi) dx$$

- (v) Show that $\pi = \frac{1}{2} \int_0^2 2\cos^{-1}(1-x) dx$

2

End of paper

1a) $A = (-1, 4)$ $B(5, -3)$

external ratio $-1:3$

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{-1 \times 5 + 3 \times -1}{-1+3}, \frac{-1 \times -3 + 3 \times 4}{-1+3} \right)$$

$$= (-4, 7 \frac{1}{2}) \quad \checkmark \checkmark$$

b) $\int \frac{x}{(x^2+2)^2} dx$ $u = x^2+2$

$$\frac{du}{dx} = 2x$$

$$\frac{dx}{du} = \frac{du}{2x}$$

$$= \int \frac{x}{u^2} \cdot \frac{du}{2x} \quad \checkmark \text{Calc}$$

$$= \frac{1}{2} \int u^{-2} du$$

$$= \frac{1}{2} [-u^{-1}] + c \quad \checkmark \text{Calc}$$

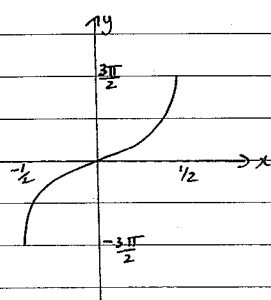
$$= \frac{-1}{2(x^2+2)} + c \quad \checkmark \text{Calc}$$

c) i) $y = 3 \sin^{-1} 2x$

$$D: -\frac{1}{2} \leq x \leq \frac{1}{2} \quad \checkmark$$

$$R: -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2} \quad \checkmark$$

ii)



d) $\frac{2x}{x-2} \leq 3$

$$2x(x-2) \leq 3(x-2)^2$$

$$2x^2 - 4x \leq 3x^2 - 12x + 12$$

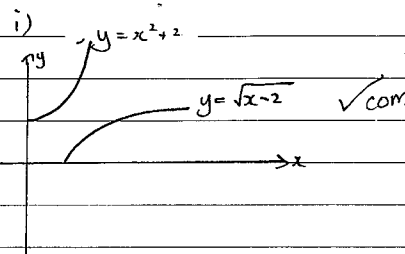
$$x^2 - 8x + 12 \geq 0$$

$$(x-6)(x-2) \geq 0$$

$$x \leq 2, x \geq 6$$

$$\checkmark \quad \checkmark$$

e) i)



ii) It passes the horizontal line test. $\checkmark \text{com}$

iii) $f^{-1}: x = \sqrt{y-2}$

$$x^2 = y-2$$

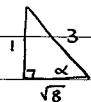
$$y = x^2 + 2 \quad \checkmark$$

iv) See graph in (i) $\checkmark \text{com}$

$$2a) \tan(\sin^{-1}(-\frac{1}{3}))$$

$$\text{Let } \sin^{-1}(-\frac{1}{3}) = \alpha$$

$$\therefore \sin \alpha = -\frac{1}{3}$$



$$\sqrt{8} = 2\sqrt{2}$$

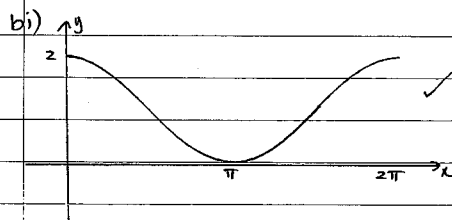
4th quad.

$$\therefore \tan(\sin^{-1}(-\frac{1}{3}))$$

$$= \tan \alpha$$

$$= \frac{-1}{2\sqrt{2}}$$

✓ Reas



$$ii) V = \pi \int_0^{3\pi/2} (1 + \cos x)^2 dx$$

$$= \pi \int_0^{3\pi/2} (1 + 2\cos x + \cos^2 x) dx$$

$$= \pi \int_0^{3\pi/2} (1 + 2\cos x + \frac{1}{2}(\cos 2x + 1)) dx$$

$$= \pi \left[x + 2\sin x + \frac{1}{2}(\frac{1}{2}\sin 2x + x) \right]_0^{3\pi/2}$$

$$= \pi \left[\frac{3\pi}{2} + 2\sin \frac{3\pi}{2} + \frac{1}{4}\sin 3\pi + \frac{3\pi}{4} \right] - 0$$

$$= \pi \left(\frac{9\pi}{4} - 2 \right)$$

✓ Calc

$$c) \int_0^1 \frac{x}{\sqrt{1-x}} dx \quad u = 1-x$$

$$\frac{du}{dx} = -1$$

$$dx = -du$$

$$\text{When } x=1 \quad u=0$$

$$x=0 \quad u=1$$

$$u=1-x \Rightarrow x=1-u$$

$$\int_1^0 \frac{1-u}{\sqrt{u}} (-du) \quad \checkmark \checkmark \text{ Calc}$$

$$\int_0^1 (u^{-1/2} - u^{1/2}) du \quad \checkmark \text{ Calc}$$

$$= \left[2u^{1/2} - \frac{2}{3}u^{3/2} \right]_0^1$$

$$= \left(2(1) - \frac{2}{3}(1) \right) - 0$$

$$= 1\frac{1}{3}$$

✓ Calc

$$d) \frac{d}{dx} \left((1-x^2)^{1/2} + x \sin^{-1} x \right)$$

Reas

$$= \frac{1}{2}(1-x^2)^{-1/2} \cdot (-2x) + x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x$$

$$= \frac{-x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

$$= \sin^{-1} x$$

✓ Reas

$$ii) \int_0^{1/2} \sin^{-1} x dx$$

$$= \left[\sqrt{1-x^2} + x \sin^{-1} x \right]_0^{1/2} \quad \checkmark \text{ Reas}$$

$$= \left(\sqrt{1-\frac{1}{4}} + \frac{1}{2} \sin^{-1} \left(\frac{1}{2} \right) \right) - \left(\sqrt{1-0} + 0 \right)$$

$$= \frac{\sqrt{3}}{2} + \frac{\pi}{12} - 1$$

✓ Reas

$$3a) y = \sin^{-1} \frac{x}{2}$$

$$y' = \frac{1}{\sqrt{4-x^2}} \quad \checkmark$$

$$\text{at } x=0 \quad y' = 1/2 \quad \checkmark$$

$$x=1 \quad y' = 1/\sqrt{3} \quad \checkmark$$

$$ii) \tan \theta = \left| \frac{1/2 - 1/\sqrt{3}}{1 + \frac{1}{2} \times \frac{1}{\sqrt{3}}} \right| \quad \checkmark$$

$$= 0.060023...$$

$$\theta = 3^\circ 26' \quad \checkmark$$

$$bi) 3\sin \theta + \cos \theta = 2$$

$$3\left(\frac{2t}{1+t^2}\right) + \left(\frac{1-t^2}{1+t^2}\right) = 2 \quad \checkmark$$

$$6t + 1 - t^2 = 2 + 2t^2 \quad \checkmark$$

$$3t^2 - 6t + 1 = 0$$

$$bii) 3t^2 - 6t + 1 = 0$$

$$t = \frac{6 \pm \sqrt{36 - 4 \times 3 \times 1}}{6}$$

$$= \frac{6 \pm \sqrt{24}}{6}$$

$$= 1.8164..., 0.1835...$$

$$\therefore \tan \frac{\theta}{2} = 1.8164...$$

$$\tan \frac{\theta}{2} = 0.1835...$$

$$\frac{\theta}{2} = 61^\circ 10'$$

$$\frac{\theta}{2} = 10^\circ 24'$$

$$\theta = 122^\circ 20'$$

$$\theta = 20^\circ 48'$$

$$\text{Test } \theta = 180^\circ$$

$$3\sin 180^\circ + \cos 180^\circ$$

$$= -1$$

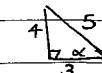
$$\therefore \theta = 180^\circ \text{ is not a solution}$$

$$\therefore \theta = 20^\circ 48', 122^\circ 20'$$

$$c) \cos^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(-\frac{3}{4}\right) = \frac{\pi}{2}$$

$$\text{Let } \cos^{-1}\left(\frac{3}{5}\right) = \alpha$$

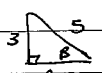
$$\text{then } \cos \alpha = \frac{3}{5}$$



1st quad

$$\text{Let } \tan^{-1}\left(-\frac{3}{4}\right) = \beta$$

$$\tan \beta = -\frac{3}{4}$$



4th quad

$$\sin(\alpha - \beta)$$

$$= \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \frac{4}{5} \times \frac{4}{5} - \frac{3}{5} \times -\frac{3}{5}$$

$$= 1 \quad \checkmark \text{ Reas}$$

$$\sin(\alpha - \beta) = 1$$

$$\alpha - \beta = \sin^{-1}(1)$$

$$= \frac{\pi}{2} \quad \checkmark \text{ Reas}$$

$$\cos^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(-\frac{3}{4}\right) = \frac{\pi}{2}$$

$$d) \int_0^{5/2} \frac{dx}{\sqrt{25-x^2}}$$

$$= \int_0^{5/2} \frac{dx}{\sqrt{2\left(\frac{25}{2}-x^2\right)}}$$

$$= \frac{1}{\sqrt{2}} \int_0^{5/2} \frac{dx}{\sqrt{\frac{25}{2}-x^2}} \quad \checkmark \text{ Calc}$$

$$= \left[\frac{1}{\sqrt{2}} \sin^{-1} \frac{\sqrt{2}x}{5} \right]_0^{5/2} \quad \checkmark \text{ Calc}$$

$$= \frac{1}{\sqrt{2}} \sin^{-1} \frac{\sqrt{2} \times \frac{5}{2}}{5} - \frac{1}{\sqrt{2}} \sin^{-1} 0$$

$$= \frac{\pi}{4\sqrt{2}} \quad \checkmark \text{ Calc}$$

$$4a) 3\cos 2x = 2 + \sin x$$

$$3(1 - 2\sin^2 x) = 2 + \sin x$$

$$3 - 6\sin^2 x = 2 + \sin x$$

$$6\sin^2 x + \sin x - 1 = 0$$

$$(3\sin x - 1)(2\sin x + 1) = 0 \quad \checkmark \text{ Reas}$$

$$\sin x = \frac{1}{3} \quad \text{or} \quad \sin x = -\frac{1}{2}$$

$$x = \sin^{-1} \frac{1}{3} \quad x = -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}$$

if n is an integer

$$x = n\pi + (-1)^n \sin^{-1} \left(\frac{1}{3}\right) \quad \checkmark \text{ Reas}$$

$$= n\pi + (-1)^n \left(-\frac{\pi}{6}\right) \quad \checkmark \text{ Reas}$$

$$b) \int_0^{\pi/6} \frac{\cos x}{4\sin^2 x + 1} dx \quad u = \sin x$$

$$\frac{du}{dx} = \cos x$$

$$\text{when } x = \pi/6 \quad u = 1/2$$

$$x = 0 \quad u = 0$$

$$\int_0^{1/2} \frac{\cos x}{4u^2 + 1} \cdot \frac{du}{\cos x} \quad \checkmark \text{ Calc}$$

$$\int_0^{1/2} \frac{1}{4u^2 + 1} du$$

$$\frac{1}{4} \int_0^{1/2} \frac{1}{u^2 + 1/4} du \quad \checkmark \text{ Calc}$$

$$\left[\frac{1}{4} \times 2 \tan^{-1} 2u \right]_0^{1/2} \quad \checkmark \text{ Calc}$$

$$= \frac{1}{2} \tan^{-1}(1) - \frac{1}{2} \tan^{-1}(0)$$

$\checkmark \text{ Calc}$

$$= \frac{\pi}{8}$$

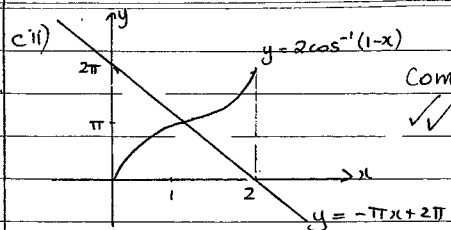
$$c) y = 2\cos^{-1}(1-x)$$

$$D: -1 \leq 1-x \leq 1$$

$$-2 \leq -x \leq 0$$

$$0 \leq x \leq 2 \quad \checkmark$$

$$R: 0 \leq y \leq 2\pi \quad \checkmark$$



iii) See sketch.

iv) $\int_0^2 2\cos^{-1}(1-x) dx$ is the area under the curve

$y = 2\cos^{-1}(1-x)$ between $x=0$ and $x=2$.

$\int_0^2 -\pi x + 2\pi dx$ is the area under the line $y = -\pi x + 2\pi$ between $x=0$ and $x=2$.

As the curve has rotational symmetry about the point $(1, \pi)$ these areas are equal. $\checkmark \text{ Com}$

$$v) \frac{1}{2} \int_0^2 2\cos^{-1}(1-x) dx$$

$$= \frac{1}{2} \int_0^2 (-\pi x + 2\pi) dx \quad \checkmark \text{ Reas}$$

$$= \frac{1}{2} \times \text{Area of } \Delta$$

$$= \frac{1}{2} \times \frac{1}{2} \times 2 \times 2\pi$$

$$= \pi \quad \checkmark \text{ Reas}$$